

CHESS DNA

Product changes since May 10

A summary of features, infra, and UX work landed in the past 4 days

Window: 2026-05-10 → 2026-05-14

Repo: Chess-dna (main)

Scope: 110 changed files · 6 new components · 3 new engine modules ·
2 new Base44 entities · ~150 new i18n keys · 25+ new diagnostic
scripts

Headline: Closed-beta gating · Replays / Time-Machine · Board-
theme catalog · Share / Achievement cards · Dedup observability

1 Closed-Beta Waitlist Gating

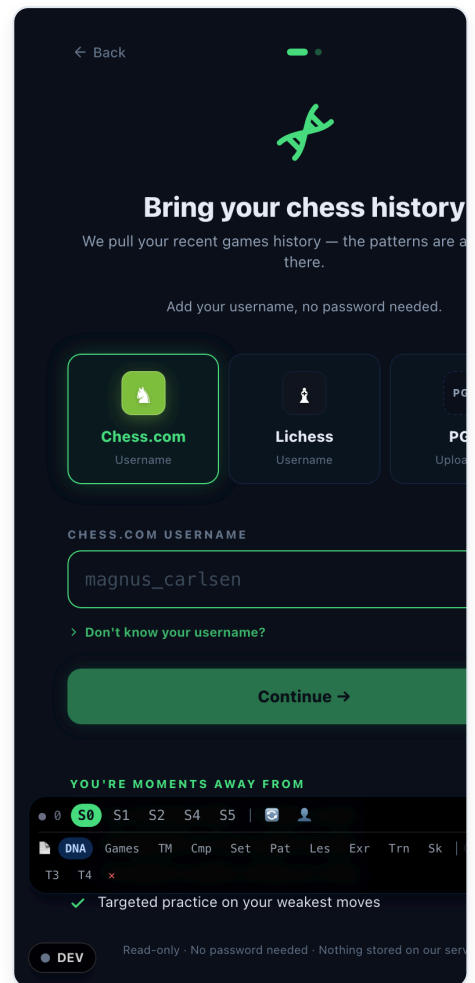
Guest mode was retired. The app now requires sign-in and checks each authenticated user against a hardcoded `BETA_TESTERS` whitelist plus a legacy-user list. Non-whitelisted users see the new **WaitlistGate** form (name, ELO, preferred mobile platform), which writes a record to the new Base44 `BetaWaitlist` entity. The `AuthContext` now tracks `betaStatus` ('pending' | 'allowed' | 'denied' | 'unknown'), with JWT-claim decoding as a fast path so whitelisted users see zero gate flicker.

NEW COMPONENTS & INFRA

```
src/components/WaitlistGate.tsx · src/shared/beta-testers.ts · src/contexts/AuthContext.tsx (beta-status logic) · src/components/AuthGuard.tsx · base44/entities/beta-a-tester.jsonc · base44/entities/beta-waitlist.jsonc · scripts/seed-beta-testers.ts
```

Route: /

Visible to: non-whitelisted signed-in users



Whitelisted users skip the gate and land in the new onboarding flow.

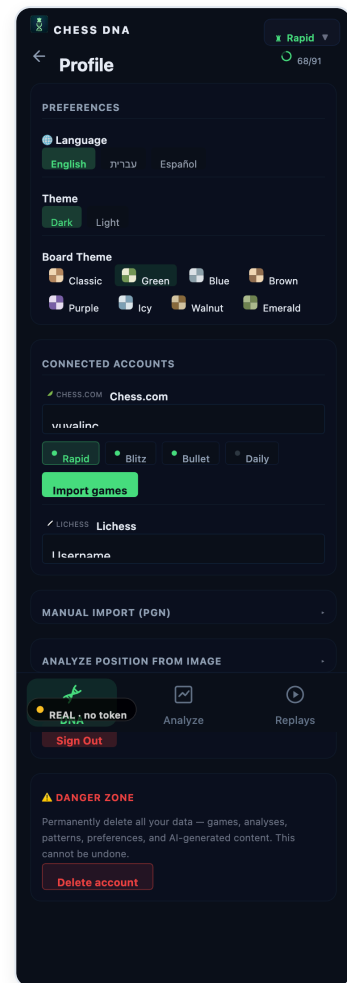
2 Board Themes & Piece Sets

A new `theme-catalog.ts` defines a controlled vocabulary of 60+ chess themes (tactical motifs, mate patterns, pawn moves, attack patterns, endgame techniques, special positions). Settings now exposes a Board Theme picker with eight themes (*Classic, Green, Blue, Brown, Purple, Icy, Walnut, Emerald*). **ThemedChessboard** accepts a `forceThemeId` override prop used by Hybrid Salvage replays to lock the classic wood look. New **cburnett-shadow-pieces** renders SVG pieces with drop shadows; piece scale increased from 0.88 → 0.97 with a 1.5× lift on drag. The new **ThemeChip** component renders inline references to chess themes (fork, pin, back-rank mate, ...) with tappable tooltips showing one-sentence i18n definitions.

FILES

```
src/shared/theme-catalog.ts · src/components/ThemeChip.tsx · src/components/cburnett-shadow-pieces.tsx · src/components/ThemedChessboard.tsx · src/components/board-themes.ts
```

Routes: `/settings` · `/games/:id` (chips in explanations)



Settings — Language, Theme, and the new **Board Theme** picker.

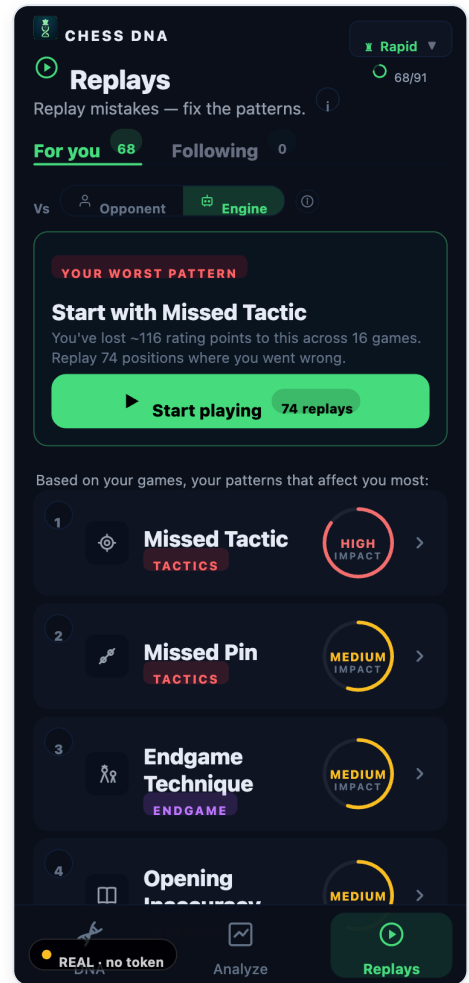
3 Replays (formerly Time Machine) + Opening-Trap Detector + Bot

The **Replays** page (renamed from "Time Machine") lets a player replay their mistakes grouped by tactical pattern. A "Worst Pattern" card surfaces the costliest leak — e.g. *"Start with Missed Tactic — you've lost ~116 rating points across 16 games"* — with a single CTA ("Start playing 74 replays"). A new `bot-mover.ts` engine offers two modes: *engine* (Stockfish at full strength) and *opponent* (replay original opponent moves on-script, fall back to Stockfish throttled to opponent's rating off-script). New `trap-detector.ts` + `opening-traps.ts` scan games for 11 well-known opening traps (Wayward Queen, Fried Liver, Englund Gambit, Halloween Gambit, ...). Play counts are tracked per challenge (legacy boolean 'checked' map migrated to numeric counts).

FILES

```
src/engine/bot-mover.ts · src/patterns/trap-detector.ts  
· src/shared/data/opening-traps.ts · src/pages/TimeMachine.tsx  
· src/hooks/useTimeMachineChallenge.ts · 30+ new tm_* i18n keys
```

Route: /timemachine



Replays — worst-pattern callout, Engine / Opponent toggle, list of patterns with impact tags.

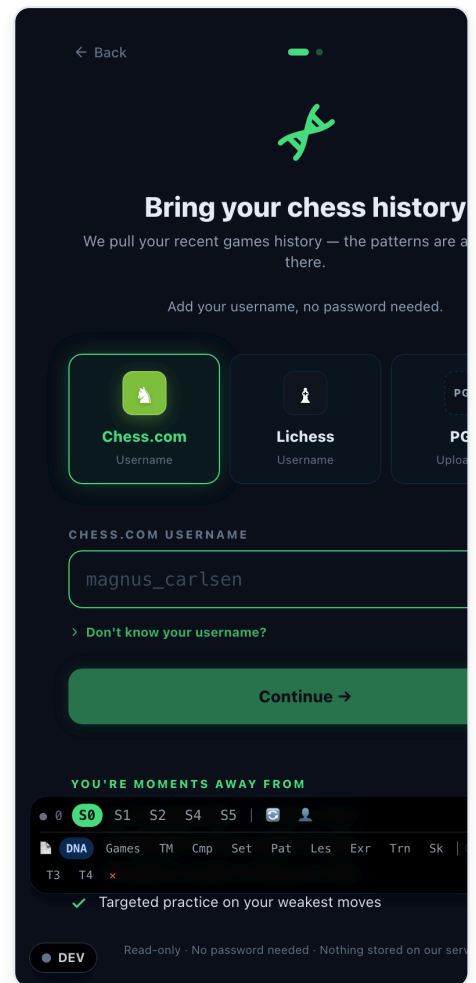
4 Onboarding & Tutorial Flow

The old stage system was replaced by a structured 5-screen sequence: **LandingScreen** (language pills + "Get Started"), **ConnectScreen** (chess.com / lichess / PGN tabs), **DecodingScreen** (analysis progress), **UnlockScreen** (skill-radar preview), **RadarRevealScreen** (final reveal). Language selection uses pills at the top right; settings are debounced and persisted to localStorage. The ConnectScreen surfaces three import sources side-by-side and a "you're moments away from" benefits list.

FILES

```
src/components/OnboardingFlow.tsx · src/pages/Overview.tsx (onboarding integration) · src/i18n/locales/en.ts (copy)
```

Route: / (first-run only)



Onboarding ConnectScreen — chess.com / lichess / PGN tabs.

5 Share Cards & Achievement Cards

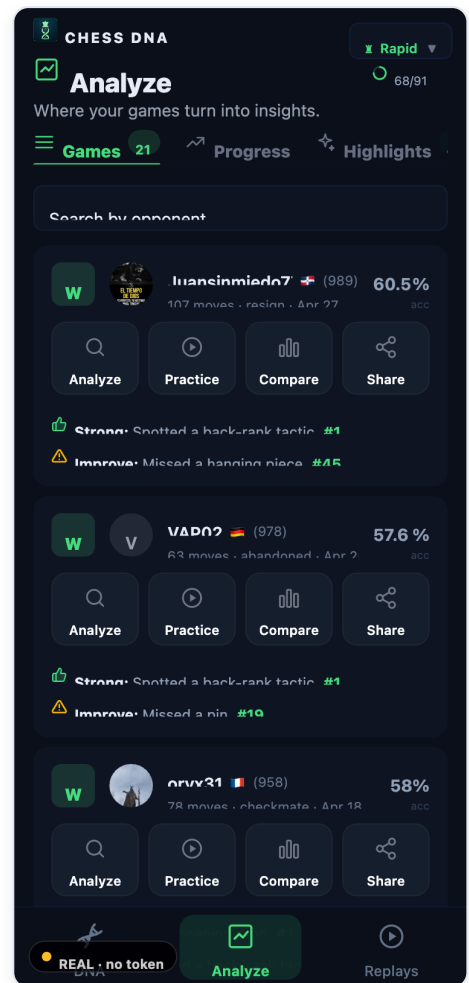
AchievementShareCard renders a dedicated 1080×1920 Instagram-Story-sized card for social sharing. The component hard-codes all dimensions and colors (no Tailwind clamp/truncate) so `html2canvas v1` captures it correctly without text clipping or overlap. `share-achievement-canvas.ts` generates the PNG from the React component. Overlay components (**MoveHighlightOverlay**, **SequenceHighlightOverlay**) annotate shared game replays with move highlights and sequence labels. Navigation copy changed: "Share your DNA" → "Share DNA".

FILES

```
src/components/share/AchievementShareCard.tsx · src/ut  
ils/share-achievement-canvas.ts · src/components/share/P  
layerCardShare.tsx · src/components/share/overlays/{Mov  
e,Sequence}HighlightOverlay.tsx · src/pages/RecentGame  
s.tsx
```

Route: /games (carousel)

Route: / (Share DNA card)



Recent Games — share carousel and achievement cards now surface here.

6 Skill Highlights & Time Window Tabs

The Overview now leads with the Skill Radar (eight-dimension polygon), the user's ELO, distance to next tier ("8 to Queen"), and an overall score. New **TimeWindowTabs** let the user switch the radar between *Last Week*, *Last Month*, and *All Time* windows (replacing "Today" / "This Week"). Below the radar, three primary actions stand out — *Share DNA*, *Your Progress*, *Profile* — followed by the prominent *Replay your mistakes* CTA that drops users straight into the new Replays flow.

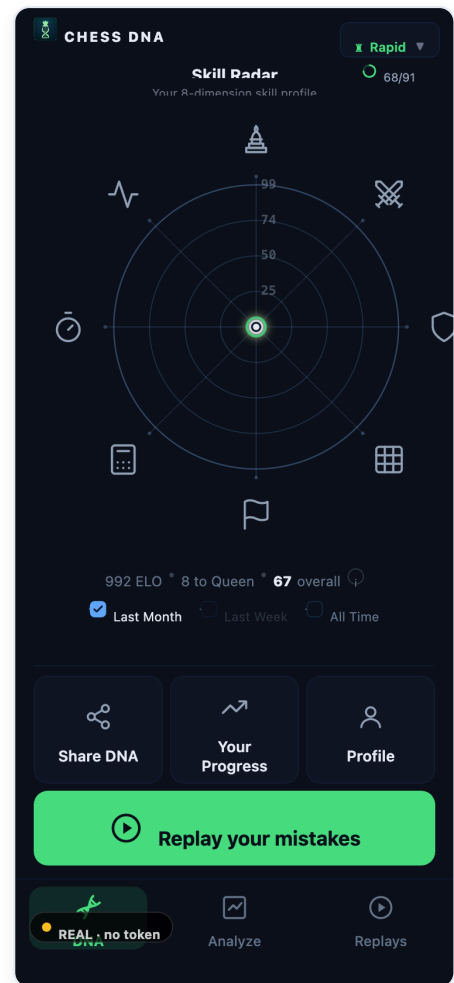
SkillHighlights shows three small cards (Strongest / Weakness / Improving) once enough trend data exists;

ExplanationText + `explanation-cache.ts` render and cache AI explanations with embedded theme chips.

FILES

```
src/components/SkillHighlights.tsx · src/components/ExplanationText.tsx · src/storage/explanation-cache.ts · src/components/TimeWindowTabs.tsx · src/pages/Overview.tsx
```

Route: / (Overview)



Overview — Skill Radar, time-window tabs, and the new *Replay-your-mistakes* CTA.

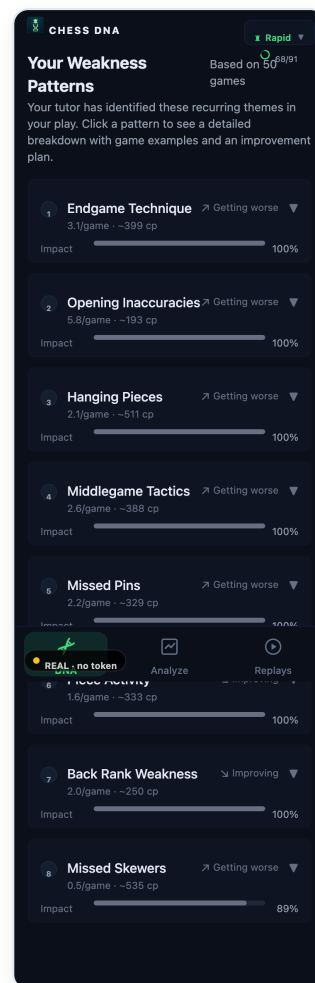
7 Patterns Page — Weakness Surface

The Patterns page now lists each weakness pattern (Endgame Technique, Opening Inaccuracies, Hanging Pieces, Middlegame Tactics, Missed Pins, Back Rank Weakness, Missed Skewers, ...) with a per-pattern impact bar (3.1/game · -399 cp) and a trend indicator (↗ *Getting worse* / *Improving*). Each pattern is tappable to drill into a detailed breakdown with examples and an improvement plan. Counts are computed from the windowed profile and refresh as new analyses complete.

FILES

```
src/pages/Patterns.tsx · src/patterns/pattern-engine.ts
· src/patterns/windowed-profile.ts · src/components/SkillRadar.tsx
```

Route: /patterns



Patterns — weakness list with impact bars and trend deltas.

8 Chess.com Sync & Dedup Diagnostics

`useChessComSync.ts` now supports guest mode (writes to `localStorage` instead of `Base44`) and tracks a `sinceMs` watermark to avoid re-polling for already-synced games. Polling pauses when the tab is hidden and resumes on `visibilitychange`. New `dedup-diagnostics.ts` maintains a 200-event ring buffer in `localStorage` so `WebView` debugging is possible without `DevTools` — each event records source (`chess.com` / `lichess`), username, gameId, outcome (`'created'` | `'existed'` | `'batch-dupe'` | `'filter-error'` | `'create-error'` | ...), and error message. `DedupDiagnosticsPanel` displays the log inside Settings. Twenty-five diagnostic scripts (`check-dvir-*`, `diagnose-*`, `cleanup-duplicate-singletons.ts`, `survey-duplicates.ts`) scan game counts, sync state, and dedup correctness; `seed-beta-testers.ts` hydrates the new `BetaTester` entity from the hardcoded list.

FILES

```
src/hooks/useChessComSync.ts · src/utils/dedup-diagnostics.ts · src/components/DedupDiagnosticsPanel.tsx · scripts/{check-dvir-*,diagnose-*,cleanup-*,survey-*.ts} · src/utils/sync-beta-testers.ts
```

Route: `/settings` (admin panel)

9 Engine & Analysis Enhancements

`game-analyzer.ts` now calls `deriveAdditionalMotifs()` after the main tactical-detector pass. `tactical-detector.ts` exports the new helper, which cheaply detects: promotion / under-promotion (from UCI suffix), en passant (pawn capture to empty square), castling (`O-O` / `O-O-O SAN`), mate-in-1 through mate-in-5 (from `evalAfter` mate scores), back-rank mate, smothered mate, mate threat, exposed king, and double check. All motifs are merged with engine-detected ones and deduped. The `MOTIF_TO_THEME` map in `theme-catalog.ts` translates engine motifs into narrative theme slugs used by the AI prompt and by the inline `ThemeChip` tooltips in explanations.

FILES

```
src/engine/game-analyzer.ts · src/engine/tactical-detector.ts · src/shared/theme-catalog.ts (MOTIF_TO_THEME)
```

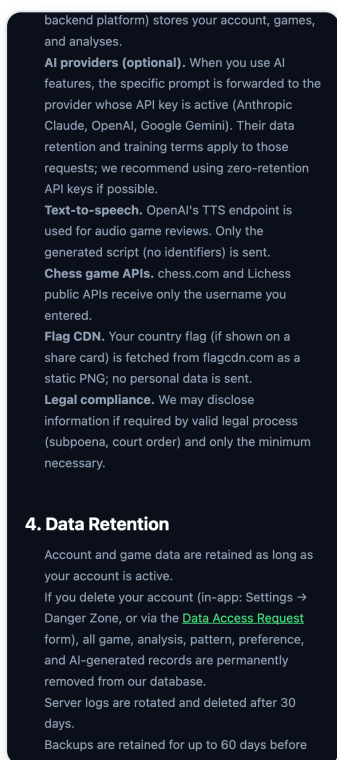
Route: `/games/:id` (move analyses)

10 I18n Expansion & Page-Level UX

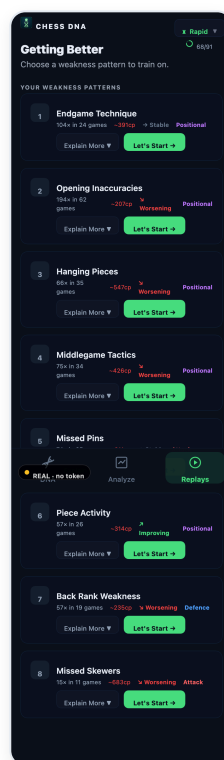
The English locale grew by ~150 keys: 60 new theme definitions (`theme_fork` / `themedef_fork` , ...), full game-termination reasons (checkmate, stalemate, time, resignation, repetition, insufficient material, 50-move rule, abandoned, rules violation), the Replays page (`tm_skip` , `tm_show_best_move` , `tm_hint` , `tm_move_progress` , `tm_worst_pattern_label` , ...), and onboarding callouts. Spanish and Hebrew were updated in parallel. Navigation labels changed: "Games" → "Analyze", "Time Machine" → "Replays"; "Try as Guest" was removed. Page-level: **Overview** dropped guest-session logic; **GameDetail** adds "You · White" / "You · Black" perspective labels; **RecentGames** surfaces the AchievementShareCard carousel; **Settings** dropped OpenAI-voice preview and TTS-model selection; **PrivacyPolicy** was updated for closed-beta terms. Stats tabs renamed: "Today" / "This Week" → "Last Week" / "Last Month".

FILES

```
src/i18n/locales/{en,es,he}.ts · src/pages/{Overview,GameDetail,RecentGames,Settings,PrivacyPolicy,AnalyticsAdmin}.tsx · src/components/AuthGuard.tsx
```



Privacy Policy — updated for closed beta.



Training page — pulls patterns into a structured improvement plan.

Summary stats

- **110 changed files** across `src/`, `scripts/`, `base44/entities/`, `public/`
- **6 new components** — `WaitlistGate` · `ThemeChip` · `SkillHighlights` · `DedupDiagnosticsPanel` · `cburnett-shadow-pieces` · `AchievementShareCard`
- **3 new engine / pattern modules** — `bot-mover` · `trap-detector` · `opening-traps` (11 traps)
- **7 new infra modules** — `beta-testers` · `theme-catalog` · `explanation-cache` · `share-achievement-canvas` · `dedup-diagnostics` · `sync-beta-testers` · 2 Base44 entities (`BetaTester`, `BetaWaitlist`)
- **~150 new i18n keys** across `en/` / `es/` / `he`
- **25+ diagnostic scripts** for `sync/` / `dedup/` / user investigation
- **1 commit landed** on main: `72ab0cf` — "Add deploy-safety guard: block builds from worktrees when main is dirty"

THEME

The arc of the week was a transition from *open-access guest mode* to *closed-beta whitelist gating*, with the supporting infrastructure — dynamic `BetaWaitlist` entity, board-theme customization, enhanced analysis-motif detection, dedup observability, and the `Replays/` / opening-trap experience — that the beta cohort will exercise.